



# **VAULTOFCODES**

## **INTERNSHIP PROJECT REPORT**

### **Password Security Analyzer**

Submitted in partial fulfillment of the Internship Program

### **Submitted By**

**Sanika Patil**

### **Internship Domain**

Ethical Hacking / Cyber Security

### **Programming Language**

Python

### **Internship Organization**

VaultofCodes

## **DECLARATION**

I hereby declare that the project report titled "**Password Security Analyzer**" submitted to **VaultofCodes** is my original work.

This project has not been submitted previously for any degree, diploma, or certification.

I further declare that all sources of information used in this project have been properly acknowledged.

Date: 25<sup>th</sup> July 2026

Place: Pune

**Sanika Patil**

## **ACKNOWLEDGEMENT**

I express my sincere gratitude to **VaultofCodes** for providing me with the opportunity to complete my internship in the field of Ethical Hacking and Cyber Security.

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I would also like to thank my family and friends for motivating me during the completion of this project.

Finally, I thank everyone who directly or indirectly contributed to the successful completion of this internship project.

**Sanika Patil**

## **ABSTRACT**

The Password Security Analyzer is a Python-based application developed to evaluate the strength and security of user passwords.

Weak passwords are one of the primary reasons behind unauthorized access and cyber attacks. This project analyzes passwords using various security parameters such as length, uppercase letters, lowercase letters, numbers, special characters, entropy calculation, and common password detection.

The application categorizes passwords into Weak, Medium, or Strong and provides recommendations for improving password security.

The project aims to increase awareness about secure password creation while demonstrating practical concepts of Python programming and cyber security.

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## **INTRODUCTION**

### **1. Introduction**

In today's digital world, passwords are the first line of defense for protecting personal, financial, and organizational information. However, many users still create weak or predictable passwords, making them vulnerable to cyber attacks such as brute force attacks, dictionary attacks, credential stuffing, and password guessing.

The **Password Security Analyzer** is a Python-based application designed to help users evaluate the strength of their passwords. It examines various characteristics of a password, including its length, complexity, use of uppercase and lowercase letters, numbers, and special characters. It also estimates password entropy and identifies commonly used passwords.

Based on the analysis, the application classifies passwords into categories such as Weak, Medium, or Strong and provides recommendations for improving password security. This project demonstrates the practical application of Python programming in the field of Ethical Hacking and Cyber Security while promoting awareness about secure password creation.

## **OBJECTIVES**

### **Project Objectives**

The primary objectives of this project are:

- To analyze password strength using multiple security parameters.
- To identify weak and commonly used passwords.
- To calculate password entropy for estimating password complexity.
- To classify passwords into Weak, Medium, and Strong categories.
- To provide suggestions for creating more secure passwords.
- To enhance awareness about password security among users.
- To demonstrate the practical implementation of Python programming concepts.
- To understand real-world cyber security practices related to authentication.

## **TECHNOLOGIES USED**

### **Software Technologies**

| Technology            | Purpose                     |
|-----------------------|-----------------------------|
| Python                | Core programming language   |
| VS Code               | Code development            |
| Tkinter (if GUI used) | Graphical User Interface    |
| Regex (re module)     | Password pattern validation |
| Math Module           | Entropy calculation         |
| OS Module             | File handling (if required) |

### **Concepts Used**

- Password Strength Analysis
- Regular Expressions
- Entropy Calculation
- Secure Coding Practices
- Cyber Security Fundamentals
- Input Validation
- Conditional Statements
- Functions and Modular Programming



## **SYSTEM REQUIREMENTS**

### **Hardware Requirements**

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 500 MB free space
- Keyboard and Mouse
- Display Monitor

### **Software Requirements**

- Operating System: Windows 10 / Windows 11 / Linux
- Python 3.10 or above
- Visual Studio Code
- Command Prompt / Terminal
- Required Python Libraries (if applicable)

### **User Requirements**

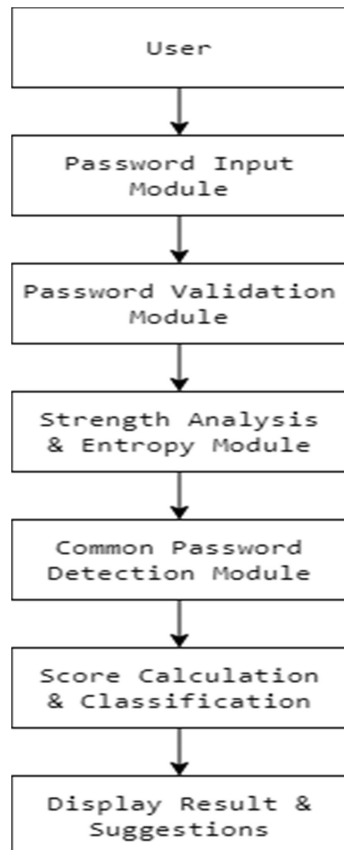
- Basic knowledge of computers
- Basic understanding of passwords and cyber security
- Python Runtime Environment installed

## PROJECT ARCHITECTURE

### System Architecture

The Password Security Analyzer follows a simple modular architecture in which the user provides a password as input. The system processes the password through different analysis modules, calculates its security score, and finally displays the password strength along with suggestions for improvement.

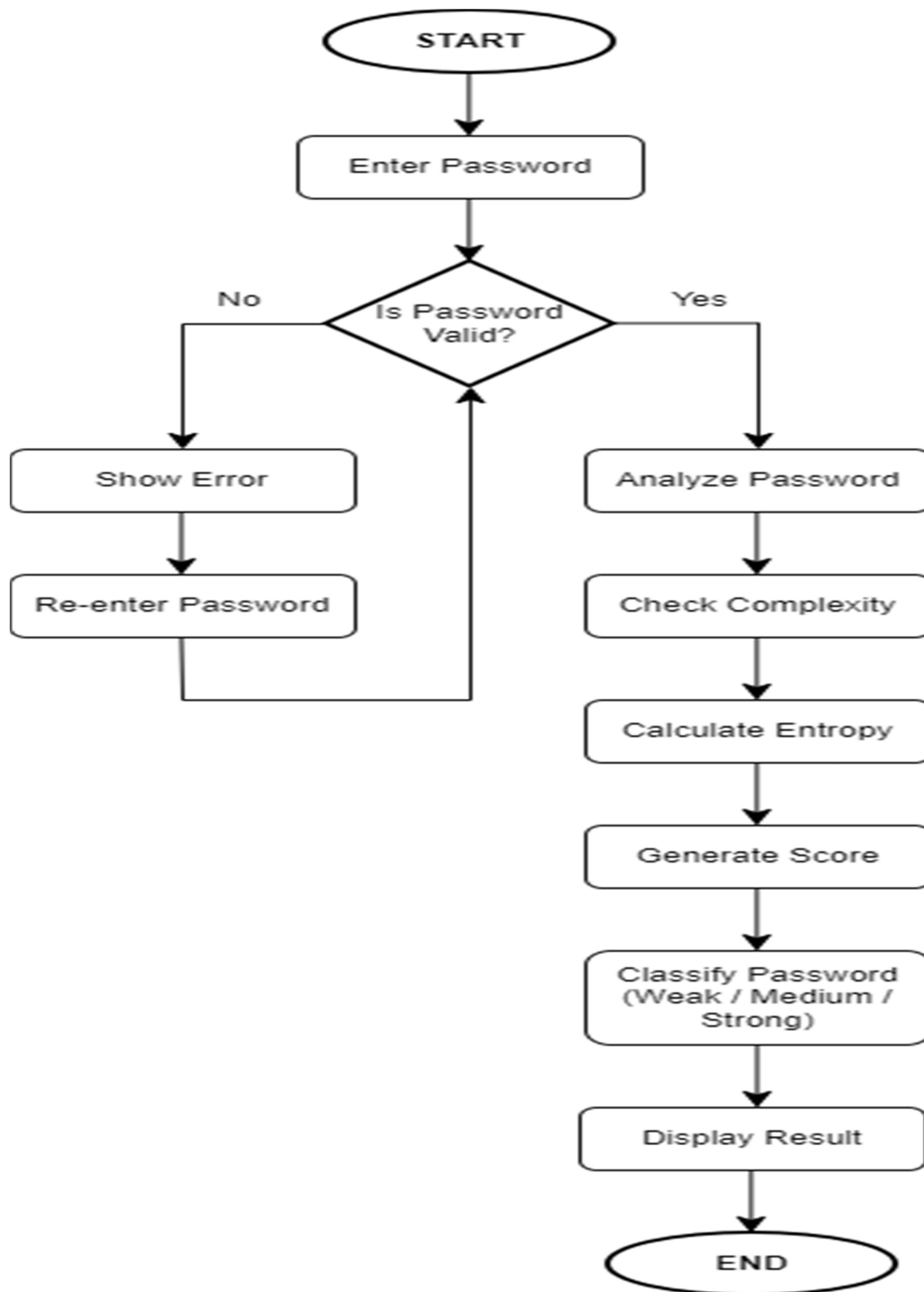
### Architecture Flow



### Description

- **Password Input Module:** Accepts the password entered by the user.
- **Validation Module:** Checks the basic validity of the password.
- **Strength Analysis Module:** Evaluates password complexity using predefined rules.
- **Entropy Module:** Calculates password entropy to estimate resistance against attacks.
- **Common Password Detection:** Detects if the password exists in a list of common passwords.
- **Result Module:** Displays the password strength and provides improvement suggestions.

## FLOWCHART



## **PROJECT MODULES**

### **Module 1 – Password Input Module**

This module allows users to enter a password for security analysis. It accepts the password and sends it to the processing modules.

#### **Functions**

- Accept password input
- Handle empty input
- Pass input for validation

### **Module 2 – Password Validation Module**

This module validates whether the password satisfies the minimum security requirements.

#### **Validation Checks**

- Minimum password length
- Maximum password length
- Empty password detection
- Invalid character handling

### **Module 3 – Password Strength Analysis**

This module evaluates the password using multiple security parameters.

#### **Parameters Checked**

- Password Length
- Uppercase Letters
- Lowercase Letters
- Numeric Digits
- Special Characters
- Repeated Characters
- Sequential Characters

Based on these parameters, the system assigns a security score.

### **Module 4 – Entropy Calculation**

Password entropy is used to estimate how difficult it would be for an attacker to guess the password.

Higher entropy indicates a stronger password and better resistance against brute-force attacks.

The entropy value is calculated based on password length and the diversity of character sets used.

## **Module 5 – Result Generation Module**

This module displays the final result after completing the analysis.

### **Output Includes**

- Password Strength
- Security Score
- Entropy Value
- Weakness Detection
- Improvement Suggestions

### **Sample Output**

Password : MySecure@123

Length : 12 Characters

Uppercase : Yes

Lowercase : Yes

Numbers : Yes

Special Characters : Yes

Entropy : High

Strength : STRONG

Security Score : 9/10

Suggestions :

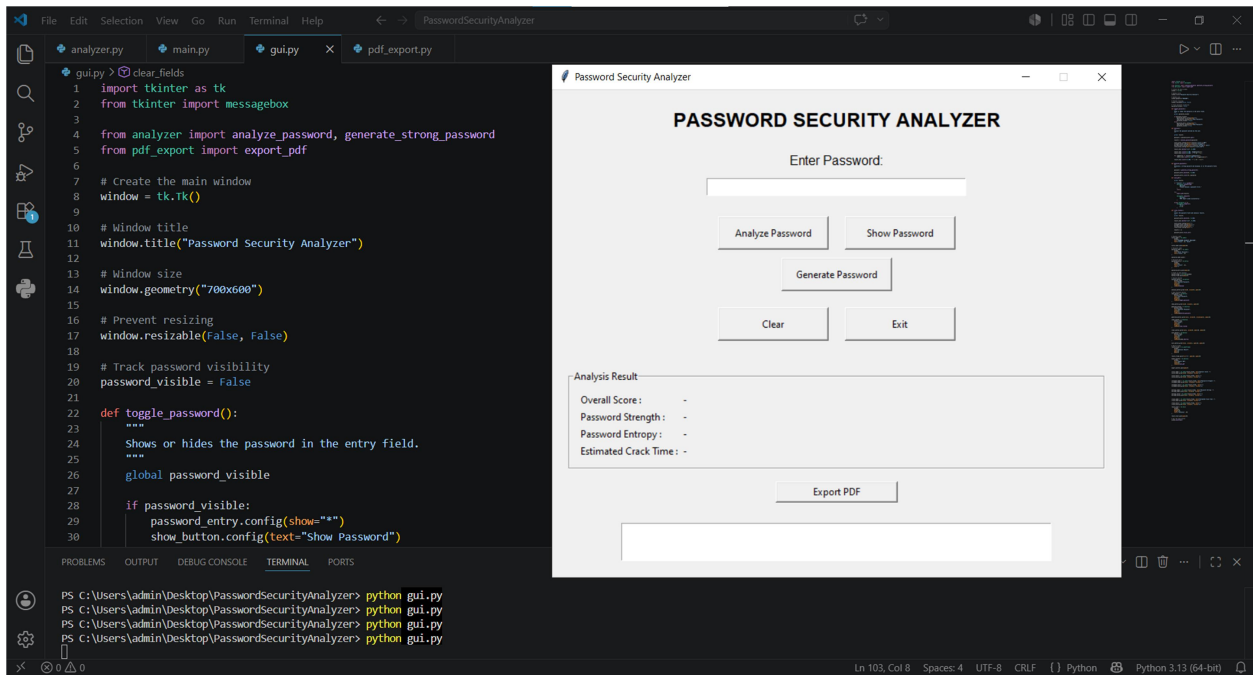
- ✓ Continue using complex passwords.
- ✓ Avoid sharing passwords.
- ✓ Enable Multi-Factor Authentication (MFA).
- ✓ Change passwords periodically.

### **Working Principle**

1. User enters a password.
2. The application validates the input.
3. The password is analyzed against security rules.
4. Entropy and strength score are calculated.
5. The password is classified as Weak, Medium, or Strong.
6. The final result and recommendations are displayed to the user.

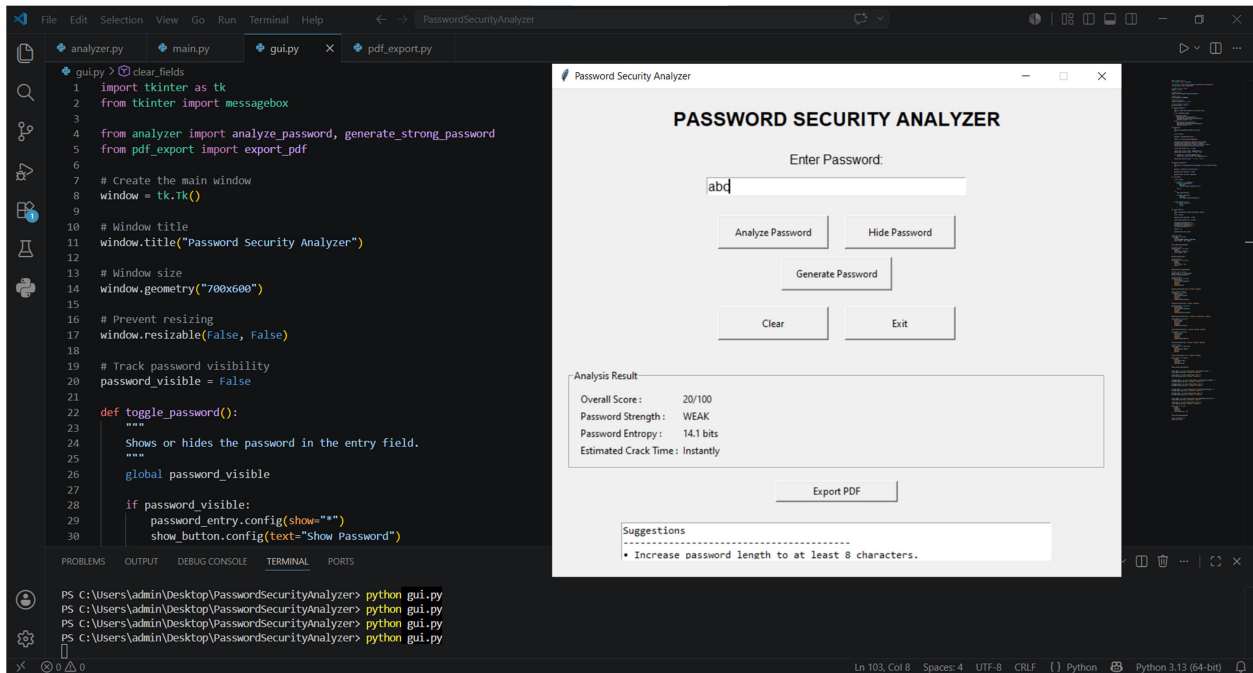
## SAMPLE OUTPUT SCREENS

### Screen 1 – Home Screen



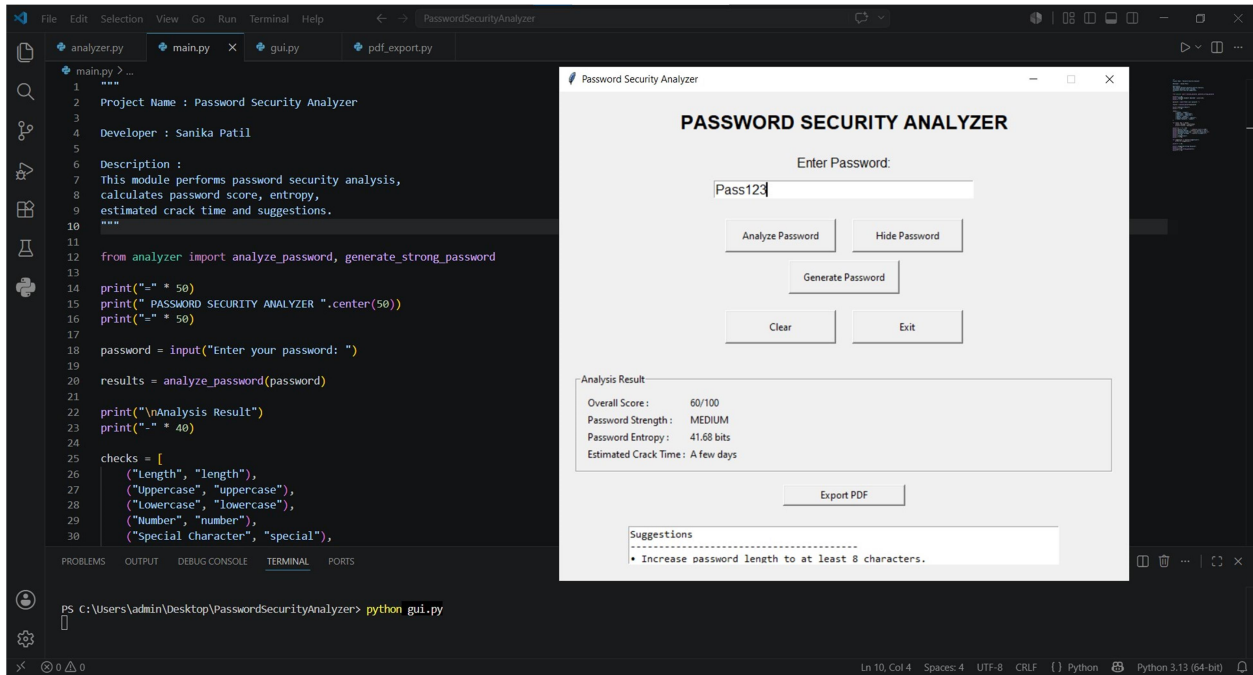
Screen 1: Home Screen of Password Security Analyzer

### Screen 2 – Weak Password Result



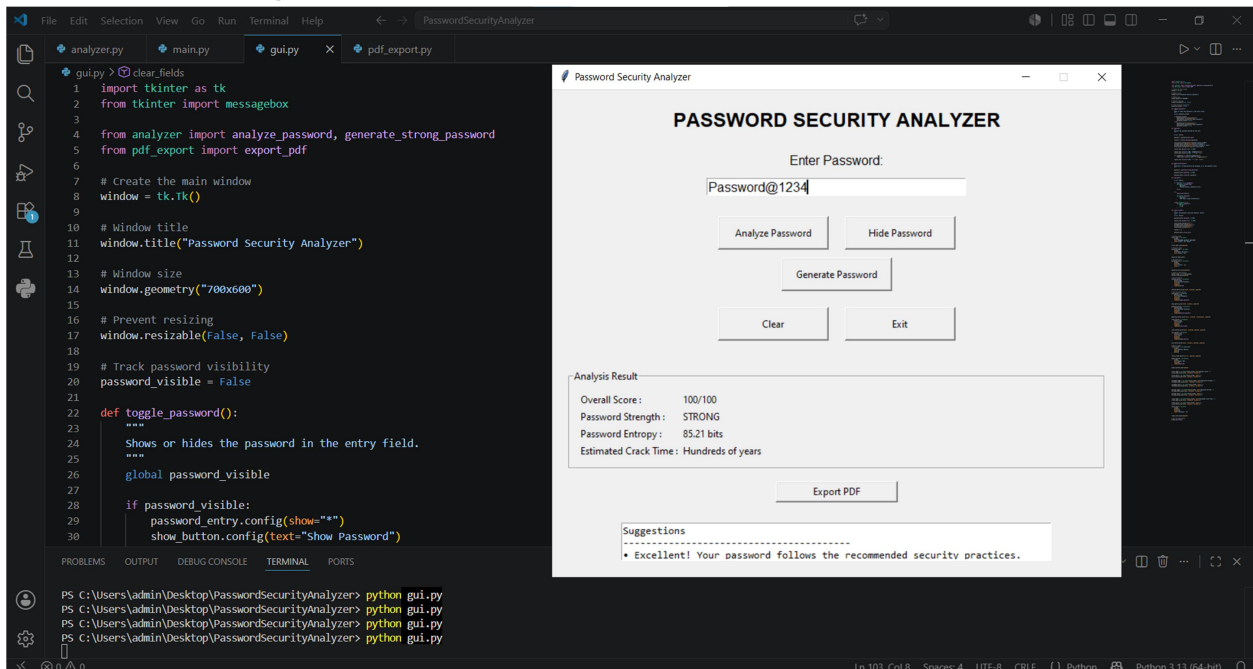
Screen 2: Weak Password Analysis

## Screen 3 – Medium Password Result



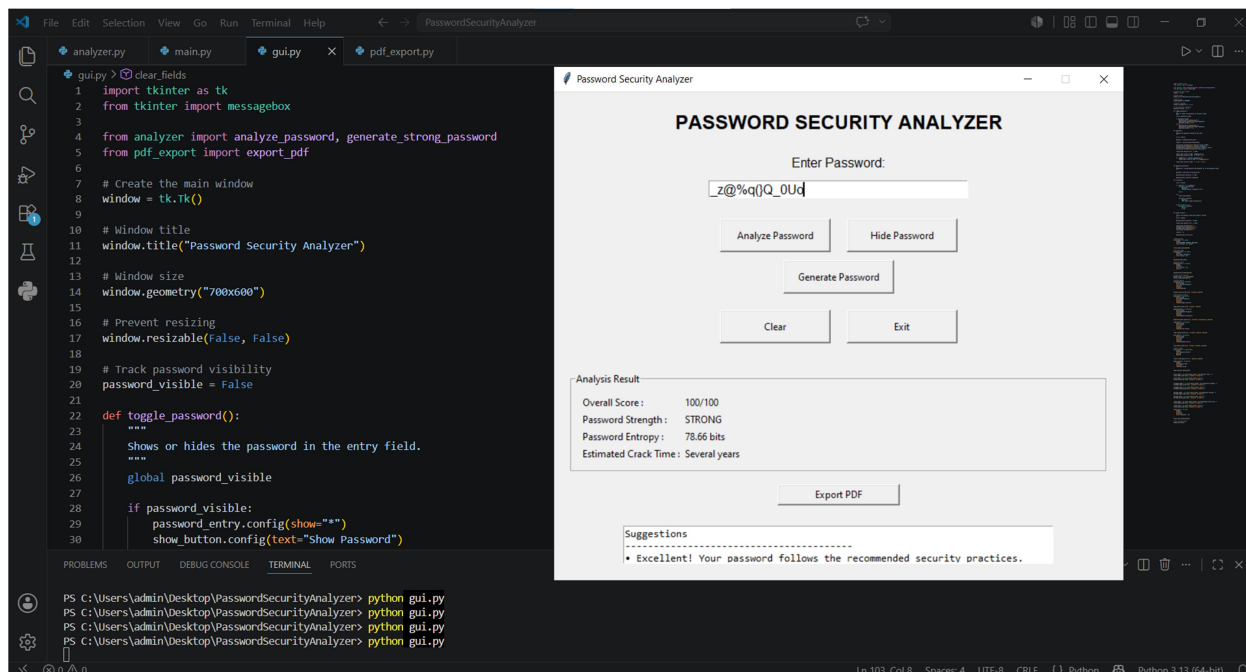
## Screen 3: Medium Password Analysis

## Screen 4 – Strong Password Result



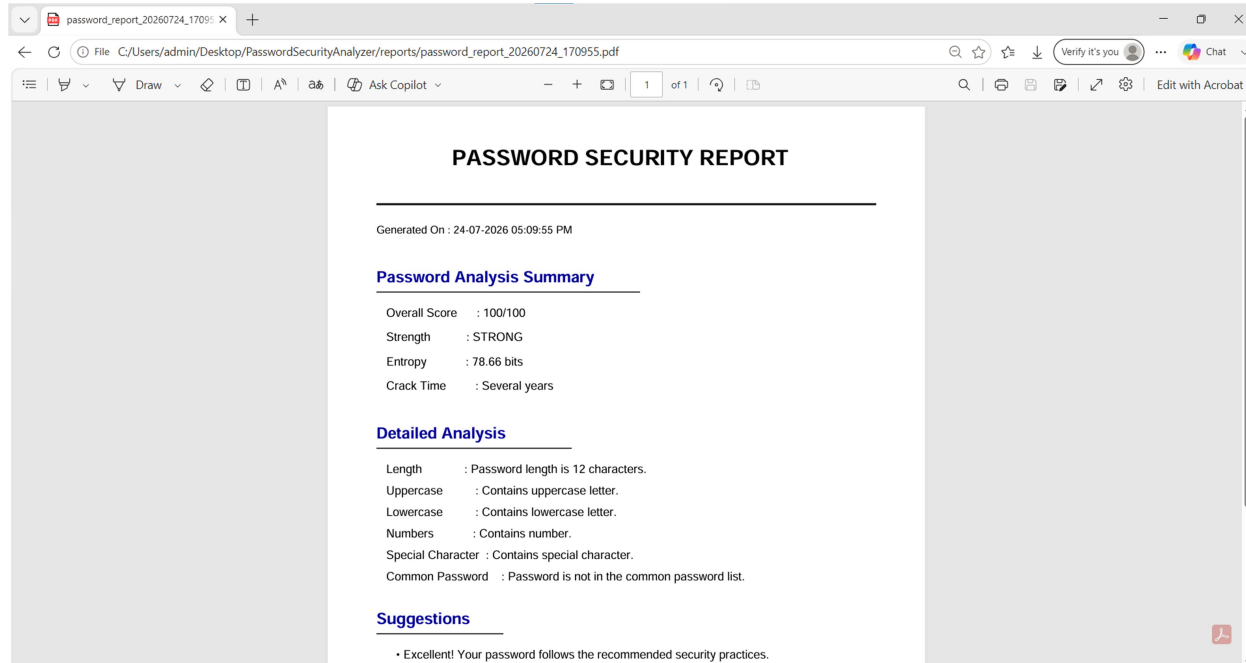
## Screen 4: Strong Password Analysis

## Screen 5 – Generated Password



Screen 5: Generated Secure Password Analysis

## Screen 6 – Generated PDF Security Report



Screen 6: Generated PDF Security Report



## **TESTING**

### **Test Cases**

| <b>Test Case</b> | <b>Input Password</b> | <b>Expected Result</b> | <b>Actual Result</b> | <b>Status</b> |
|------------------|-----------------------|------------------------|----------------------|---------------|
| TC-01            | 12345                 | Weak                   | Weak                 | Pass          |
| TC-02            | password              | Weak                   | Weak                 | Pass          |
| TC-03            | Password123           | Medium                 | Medium               | Pass          |
| TC-04            | MySecure@123          | Strong                 | Strong               | Pass          |
| TC-05            | Admin@2026#           | Strong                 | Strong               | Pass          |

### **Testing Summary**

- Password length validation was successful.
- Character type detection worked correctly.
- Entropy calculation produced expected values.
- Strength classification was accurate.
- Suggestions were generated based on password weaknesses.

## **ADVANTAGES**

The Password Security Analyzer provides several benefits:

- Helps users create stronger passwords.
- Detects weak and commonly used passwords.
- Reduces the risk of brute-force and dictionary attacks.
- Calculates password entropy for better security evaluation.
- Easy to use with a simple interface.
- Developed using Python, making it portable and lightweight.
- Useful for educational and cyber security awareness purposes.
- Can be integrated into authentication systems.

## **LIMITATIONS**

Although the project performs password strength analysis effectively, it has certain limitations:

- It does not verify whether a password has been exposed in real-world data breaches.
- Password analysis is based on predefined rules and scoring.
- It does not implement password hashing or encryption.
- Multi-factor authentication (MFA) is not included.
- It analyzes only one password at a time.
- The project is intended for educational purposes and not as a complete enterprise security solution.

## **FUTURE SCOPE**

The project can be enhanced with the following features:

- Integration with a database of leaked passwords.
- AI/ML-based password strength prediction.
- Password generation using secure random algorithms.
- Support for Multi-Factor Authentication (MFA).
- Integration with web applications.
- Password history management.
- Real-time security monitoring.
- Cloud-based password security analysis.
- User account management and reporting dashboard.

## **CONCLUSION**

The Password Security Analyzer successfully demonstrates the practical application of Python programming in the field of Ethical Hacking and Cyber Security. The application evaluates password strength using multiple security parameters such as password length, character diversity, entropy, and common password detection.

The project highlights the importance of strong passwords in protecting sensitive information from unauthorized access and common cyber attacks. It also provides useful recommendations to help users improve password security.

Overall, this project has enhanced practical knowledge of secure coding, password analysis techniques, and cyber security concepts while meeting the objectives of the internship.

## **REFERENCES**

The following resources were referred to during the development of this project:

1. Python Software Foundation. *Python Documentation*. <https://docs.python.org/>
2. OWASP Foundation. *Password Security Guidelines*. <https://owasp.org/>
3. National Institute of Standards and Technology (NIST). *Digital Identity Guidelines (SP 800-63B)*.
4. Microsoft Security Documentation.
5. Cybersecurity and Infrastructure Security Agency (CISA) – Password Best Practices.
6. Various Python tutorials and educational resources related to password validation and cyber security.